

Wayfair E-commerce Sales & Profitability Analysis

1. Project Overview

Wayfair operates a large e-commerce marketplace where sales performance can vary across products, categories and time periods. This analysis evaluates historical sales data from 2024-2025 to understand changes in sales and profitability, identify products and categories contributing to performance and examine the roles of returns and cancellations.

This project uses a simulated dataset based on a Wayfair business scenario and is intended to demonstrate an end-to-end data analytics workflow.

Business Questions:

1. How has sales performance changed from 2024-2025?
2. Which products and categories drive revenue and profitability?
3. How are returns and cancellations affecting performance?

2. Dataset & Methodology

2.1 Dataset Overview

This project uses four datasets covering customers, products, orders and order items. The data provides information needed to analyse sales, profitability, product performance and order status from 2024-2025.

Dataset	Records	Purpose
Customers	3,600	Customer and acquisition information
Products	90	Product, category, price and cost information
Orders	9,207	Order dates, status, payment, discounts and shipping
Order items	15,409	Product-level transaction details

2.2 Tools & Technologies

The analysis was completed using:

- **Python** – Data Understanding, Cleaning and Exploratory Analysis
- **PostgreSQL** – Data Modelling, KPI calculations and Business-question Analysis
- **Power BI** – Interactive Dashboard and Visualization

2.3 Analytical Approach

The project followed an end-to-end analytical workflow:

Raw Data → Data Cleaning → Exploratory Data Analysis → Data Modelling → KPI Definition → SQL Analysis → Power BI Dashboard → Insights & Recommendations

The analysis focused on three key areas: sales performance over time, product and category profitability and the impact of returns and cancellations.

3. Data Cleaning & Data Modelling

3.1 Data Cleaning

The raw datasets were reviewed for missing values, duplicate records, invalid values and data-type issues before analysis.

The main cleaning steps included:

- Removing **38 exact duplicate order records**.
- Removing **199 order-item records** with missing unit prices.
- Converting order and customer data fields from text to date format.
- Replacing missing discount codes with **“No Discount”**.
- Standardizing product category names by removing extra spaces and applying consistent capitalization.
- Validating relationships between customers, orders, order items and products.

After cleaning, the final datasets contained **3,600 customers, 9,207 orders, 15,409 order items and 90 products**.

3.2 Derived Financial Metrics

To support sales and profitability analysis, three financial metrics were derived from the transaction-level data:

Revenue = (Quantity x Unit Price) – Discount Amount

Cost = Quantity x Unit Cost

Profit = Revenue – Cost

These derived metrics were stored in the fact_order_items table and used consistently across the SQL analysis and Power BI dashboard.

3.3 Data Modelling

A relational data model was created in PostgreSQL to support consistent analysis and reporting. The model consists of customer, product, date and order dimensions connected to the order-item fact table.

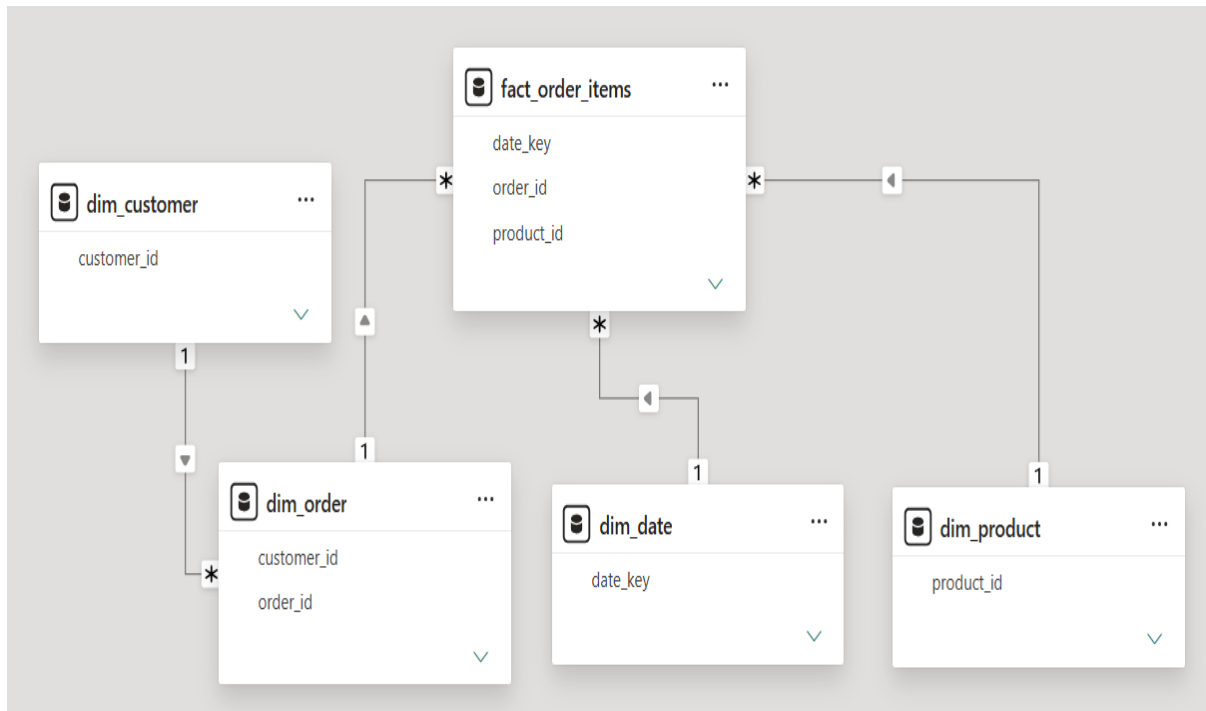


Fig 1 – Power BI data model for the Wayfair E-commerce Sales & Profitability Analysis

The fact table contains transaction-level quantities, prices, discounts, revenue, cost and profit. Dimension tables provide descriptive information used to analyse these transactions by customer, product, date and order status.

4. Analysis & Key Findings

4.1 Sales Performance Over Time

Sales performance improved from 2024 to 2025. Total revenue increased from **\$926.5K to \$1.12M**, representing **21.0% growth**, while total profit increased by **27.7%**.

Profit margin also improved from **47.75% to 50.39%**, indicating that profit grew faster than revenue during the period analysed.

Monthly analysis showed that **November was the strongest month in both years**. Revenue reached approximately **\$141.1K in November 2025**, compared with **\$107.5K in November 2024**.

4.2 Product & Category Performance

Furniture generated the highest total revenue at approximately **\$367.9K** and the highest total profit at approximately **\$170.3K**. However, its profit margin was **46.30%**, lower than several other categories.

Storage recorded the highest category profit margin at **55.59%**, showing that high revenue volume was not always associated with the highest profitability.

At the product level, **Cotton Patio Chair** generated the highest revenue and profit, with approximately **\$74.5K in revenue** and **\$33.3K in profit**.

4.3 Returns & Cancellations

The analysis identified returns and cancellations as areas requiring further attention. Across all cleaned orders, the overall return rate was **4.42%**, while the cancellation rate was **3.74%**.

Furniture had the highest return rate at **11.66%**, considerably higher than the other categories. **Bath** had the highest cancellation rate at approximately **4.50%**.

These patterns indicate that return and cancellation behaviour varies across categories and should be monitored alongside sales and profitability metrics.

4.4 Power BI Dashboard

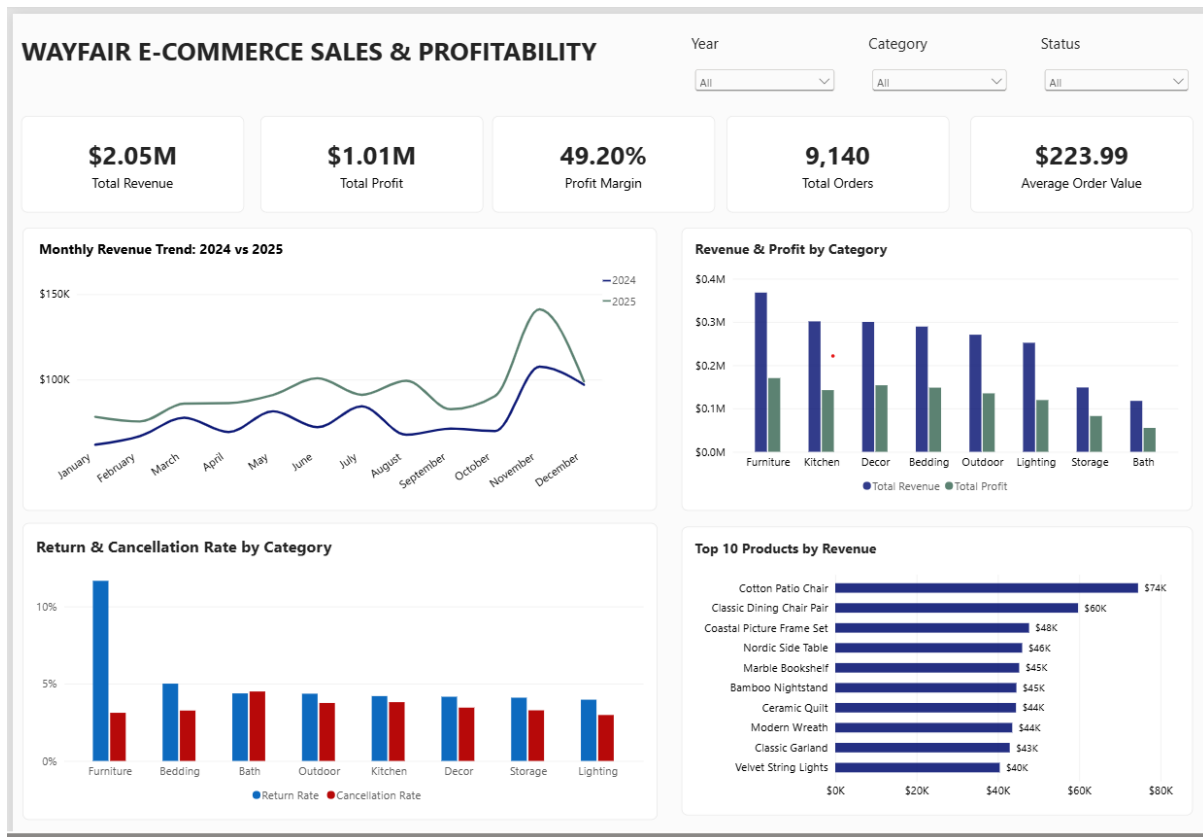


Fig 2 – Power BI Dashboard summarizing sales performance, profitability, product performance and order-status analysis

The Power BI dashboard brings together the key findings from the analysis through interactive KPIs, trend analysis, category and product performance and return and cancellations rates. Users can filter the analysis by year, category and order status to explore different aspects of business performance.

5. Business Recommendations

- **Monitor category-level profitability:** Track revenue together with profit margin to identify categories where strong sales volume does not necessarily translate into higher profitability.
- **Investigate Furniture returns:** Furniture has the highest return rate at **11.66%**. Further analysis should examine potential causes such as product expectations, product descriptions, quality or fulfilment.
- **Review Bath cancellations:** Bath has the highest cancellation rate at approximately **4.50%**. Investigating the reasons for cancellations could help identify operational or product-related issues.
- **Evaluate high-revenue products:** Products generating high revenue but comparatively lower profit margins should be reviewed for pricing, cost and promotional strategies.

- **Plan around seasonal demand:** November recorded the highest revenue and order volume in both years. Seasonal demand patterns should be considered when planning inventory and promotional activity.

6. Limitations

- The dataset represents a simulated e-commerce business scenario and does not represent internal Wayfair data.
- 199 order-item records with missing unit prices were removed, which means some transactions are not included in transactional-level revenue and profitability analysis.
- The dataset does not contain refund amounts or the financial impact of cancellations and returns, so their direct monetary impact could not be calculated.
- The analysis covers only the 2024-2025 period, so longer-term trends could not be evaluated.

7. Conclusion

This analysis evaluated e-commerce sales performance, profitability, product and category performance and order-status patterns for 2024-2025. The result showed overall improvement in revenue and profitability in 2025, with Furniture contributing the highest revenue and profit. The analysis also identified differences in return and cancellation rates across categories, highlighting areas for further investigation. The Power BI dashboard brings these findings together into an interactive view that can support ongoing monitoring of sales, profitability and order performance.